

IMC PROSPERITY · ROUND 5 · IGNITH EXCHANGE
scenarios=200,000 seed=42 budget=1,000,000
fee = (v/100)²×BUDGET | LONG+SHORT, Σ|alloc|=100 %

— ALL 9 PRODUCTS (direction EV conviction) —				
▲ LONG	Thermalite Core	EV=+45.0%	[██████████]	] 9/10
▼ SHORT	Lava Cake	EV=+42.0%	[██████████]	] 9/10
▲ LONG	Magma Ink	EV=+30.0%	[██████████]	] 8/10
▼ SHORT	Pyroflex Cell	EV=+25.0%	[██████████]	] 7/10
▲ LONG	Sulfur Ltd.	EV=+20.0%	[██████████]	] 8/10
▼ SHORT	Obsidian Cutlery	EV=+20.0%	[██████████]	] 6/10
▲ LONG	Scoria Paste	EV=+18.0%	[██████████]	] 5/10
▲ LONG	Volcanic Incense	EV=+15.0%	[██████████]	] 3/10
▼ SHORT	Ashes of Phoenix	EV=+10.0%	[██████████]	] 5/10

— ANALYTICAL KKT OPTIMUM (closed-form, unconstrained integer rounding) —

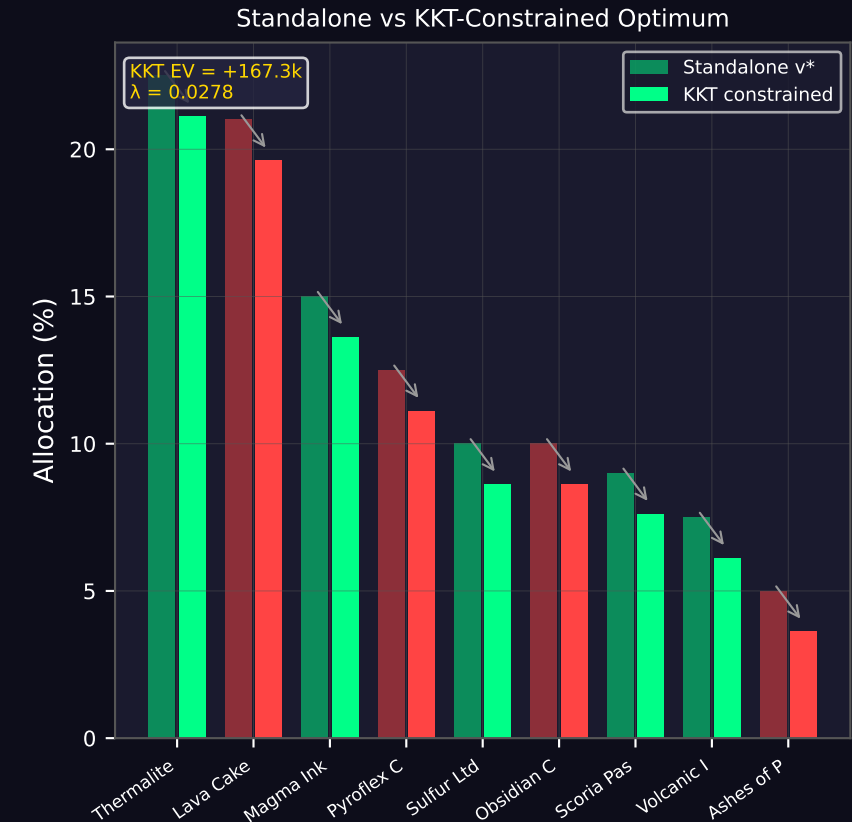
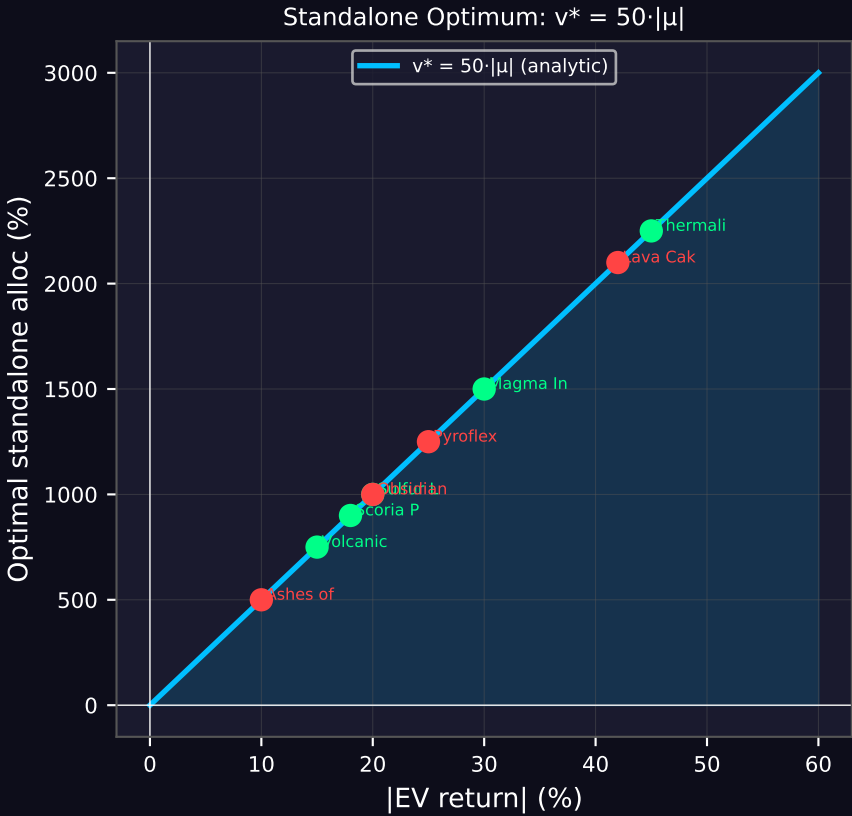
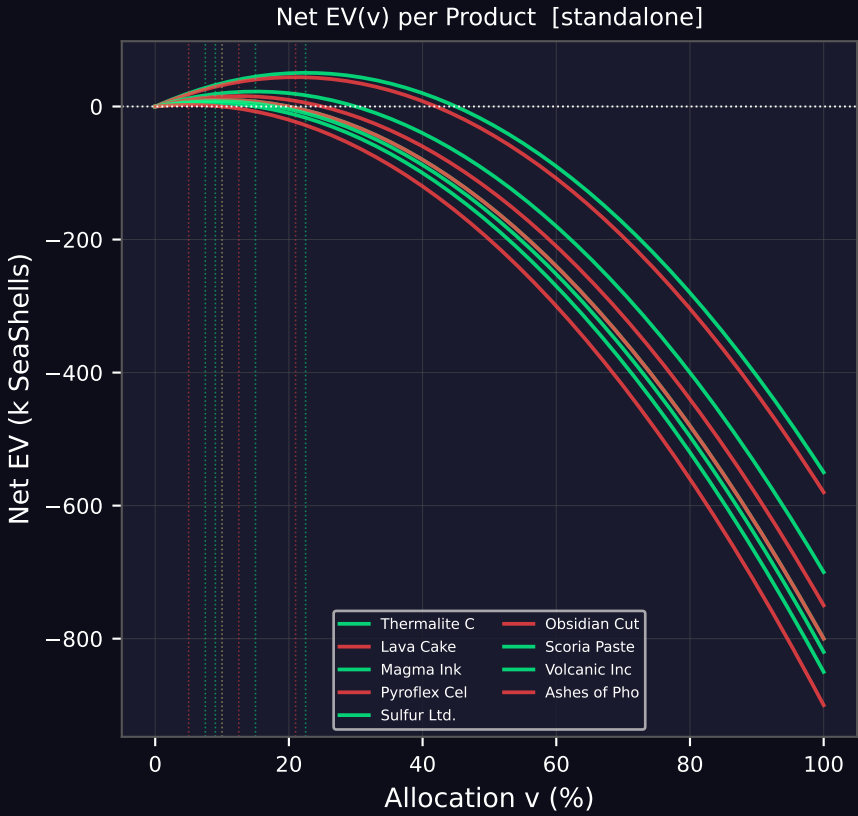
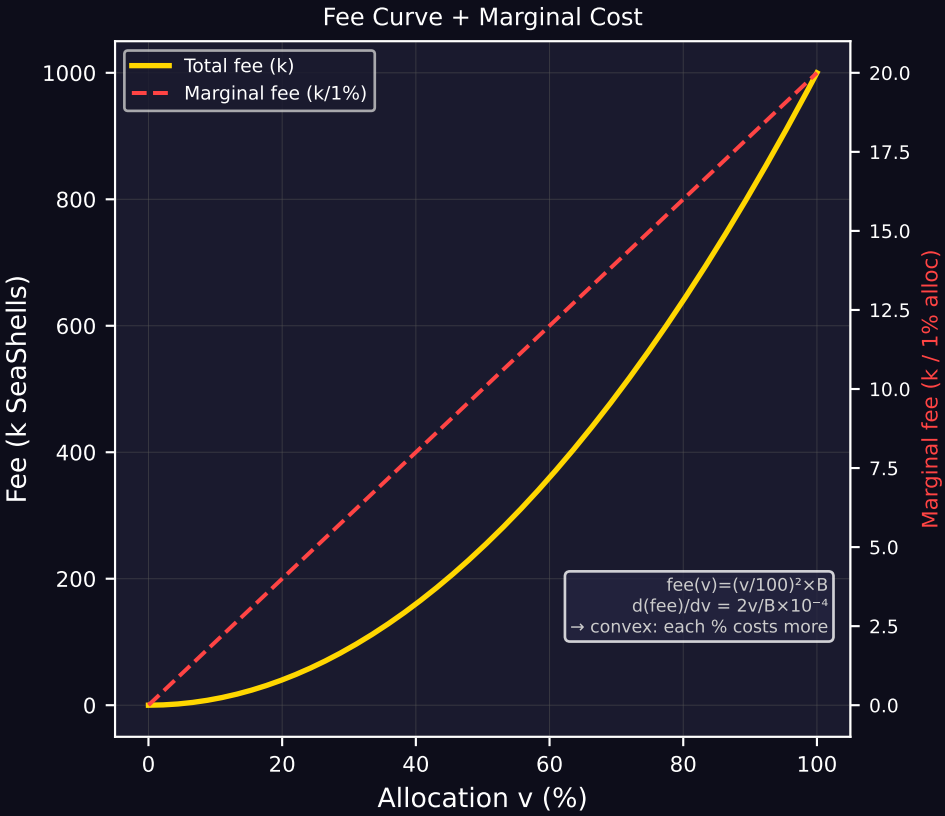
L	Thermalite Core	21%	██████████
S	Lava Cake	20%	██████████
L	Magma Ink	14%	██████████
S	Pyroflex Cell	11%	██████████
L	Sulfur Ltd.	9%	██████████
S	Obsidian Cutlery	9%	██████████
L	Scoria Paste	8%	██████████
L	Volcanic Incense	6%	██████████
S	Ashes of Phoenix	4%	██████████
Theoretical EV = +167.3k SeaShells (unconstrained analytic)			

— OPTIMIZER RESULT (Max Sharpe, Full-Eval) —

L	Thermalite Core	+20%	██████████
S	Lava Cake	-20%	██████████
L	Magma Ink	+10%	██████████
S	Pyroflex Cell	-20%	██████████
S	Obsidian Cutlery	-10%	██████████
L	Scoria Paste	+10%	██████████
S	Ashes of Phoenix	-10%	██████████

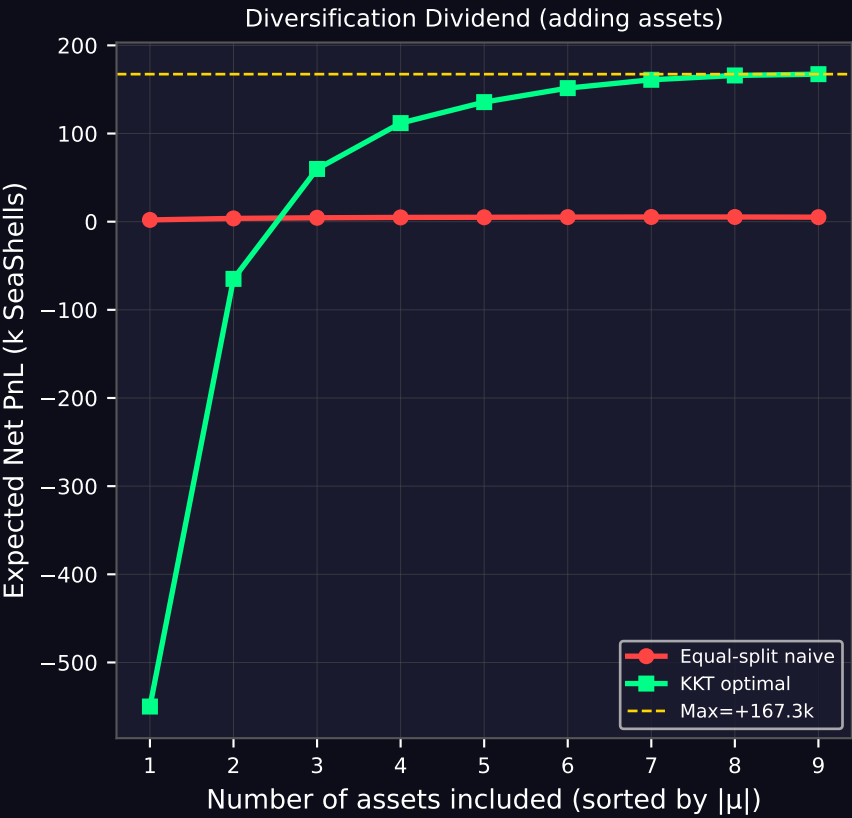
Σ|alloc|=100% Total fees = +160.0k
EV=+179.6k Sharpe=2.087 Sortino=6.845
Win Rate=98.3% VaR95=+40.7k

Fee Cost Function Analysis — $\text{fee}(v\%) = (v/100)^2 \times 1,000,000$ (quadratic, convex)



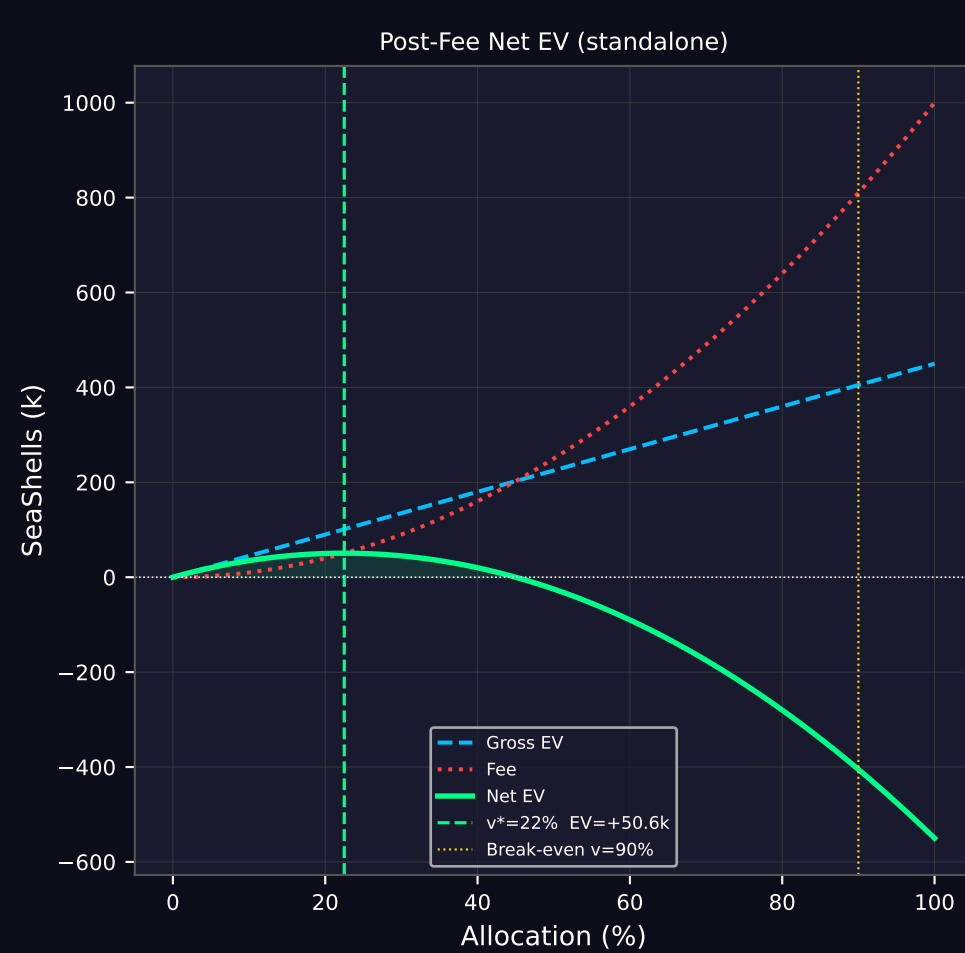
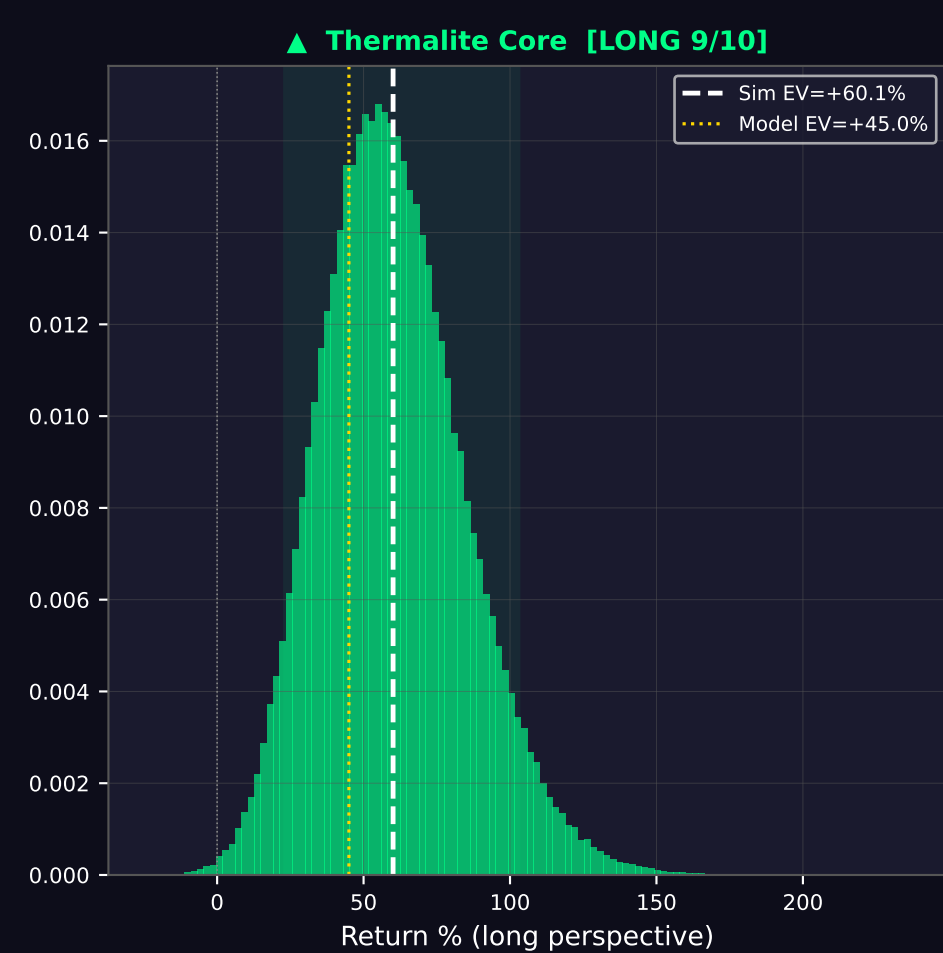
Fee Efficiency per Product at Standalone v^*

Product	Dir	$ \mu $	v^*_{SA}	v^*_{KKT}	Gross	Fee	Neteff
Thermalite C	L	45%	22.5%	21.1%	+101.2k	+50.6k	50%
Lava Cake	S	42%	21.0%	19.6%	+88.2k	+44.1k	50%
Magma Ink	L	30%	15.0%	13.6%	+45.0k	+22.5k	50%
Pyroflex Cel	S	25%	12.5%	11.1%	+31.2k	+15.6k	50%
Sulfur Ltd.	L	20%	10.0%	8.6%	+20.0k	+10.0k	50%
Obsidian Cut	S	20%	10.0%	8.6%	+20.0k	+10.0k	50%
Scoria Paste	L	18%	9.0%	7.6%	+16.2k	+8.1k	50%
Volcanic Inc	L	15%	7.5%	6.1%	+11.2k	+5.6k	50%
Ashes of Pho	S	10%	5.0%	3.6%	+5.0k	+2.5k	50%



All 9 Ignith Products — Return Distributions (path simulation)

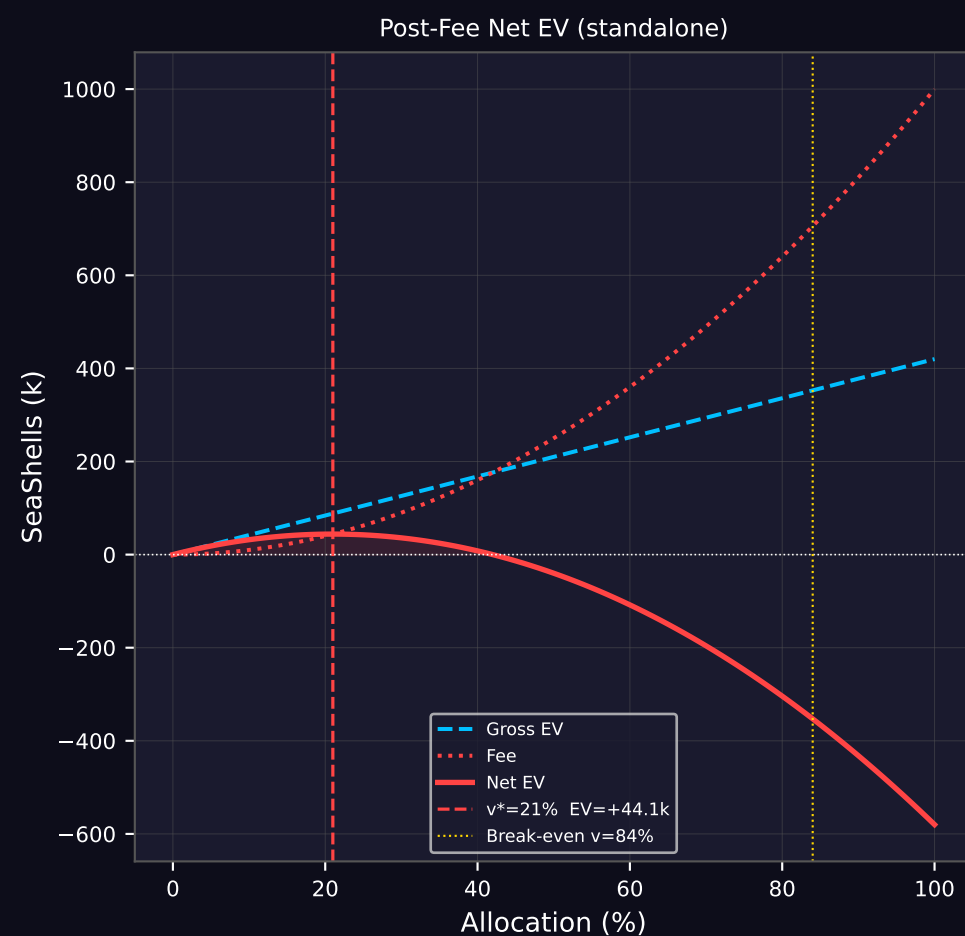
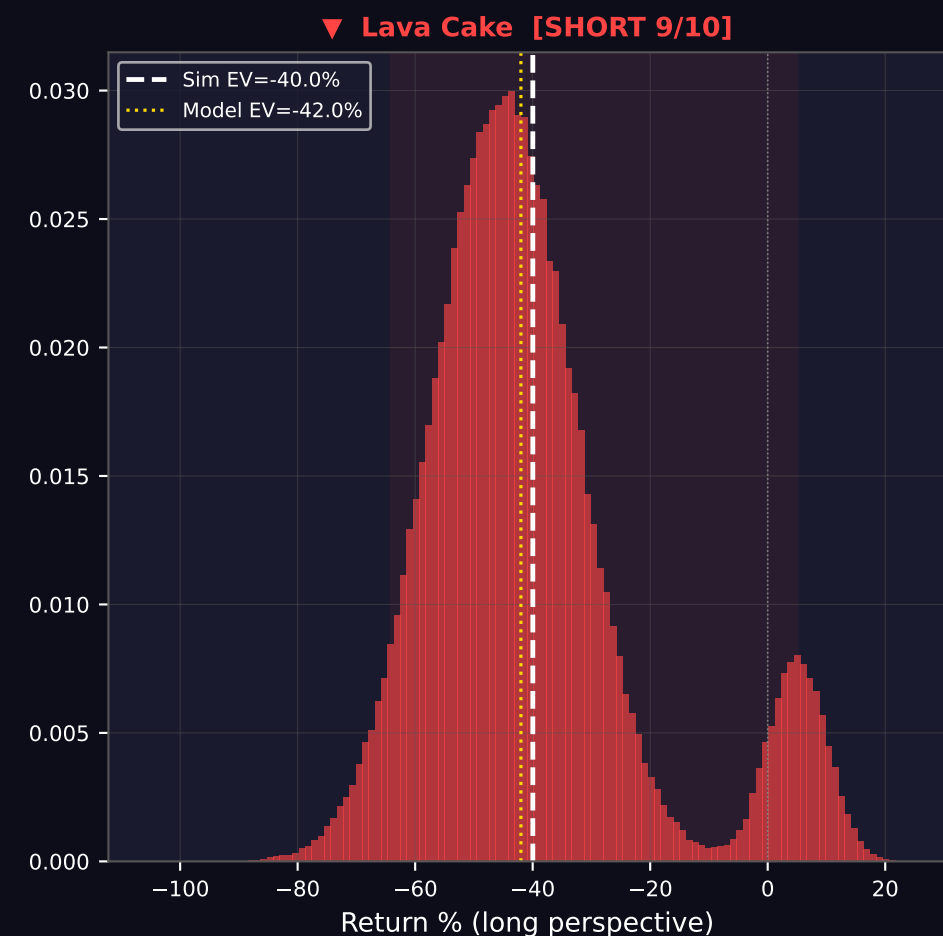




Signal: LONG ▲ [9/10]
EV (long): +45.0% EV (optimal dir): +45.0%
Std: 24.5% Model: trend_gbm
Standalone v* = 22% Max net EV = +50.6k

P1: +10.5%
P5: +23.0%
P10: +30.1%
P25: +42.9%
P50: +58.3%
P75: +75.4%
P90: +92.3%
P95: +103.1%
P99: +124.9%

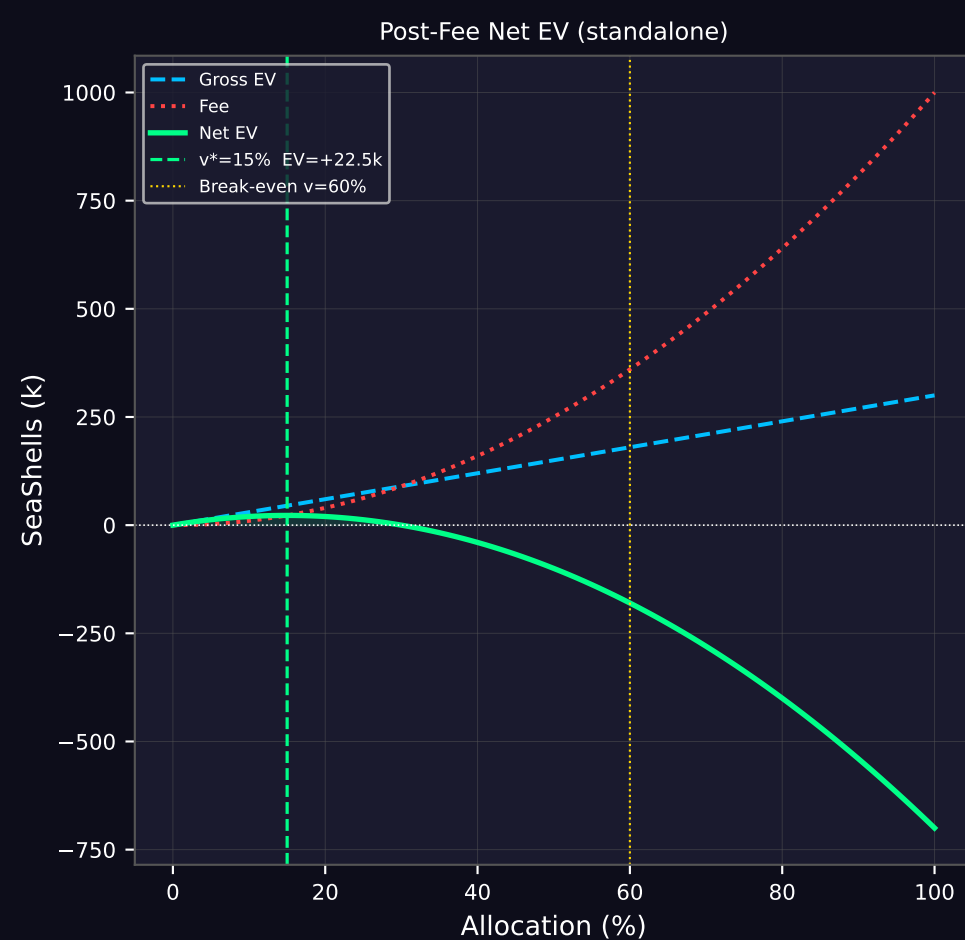
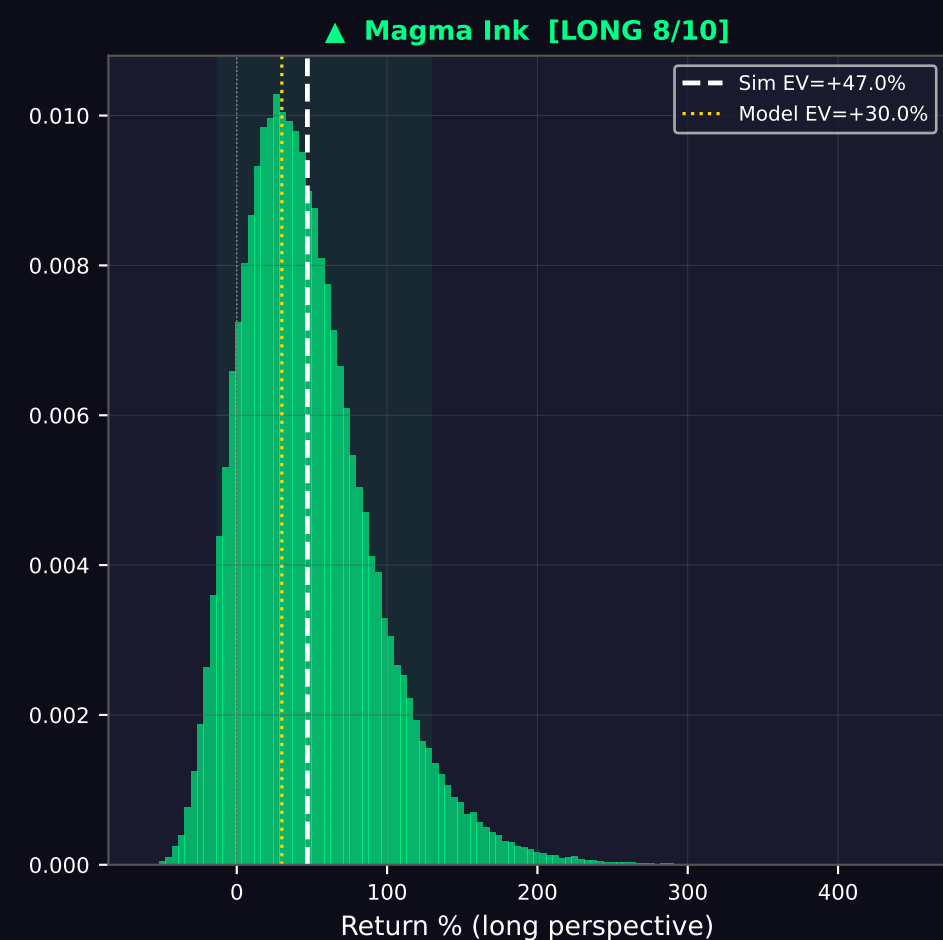
— THESIS —
Analyst forecast: users +174 % (1.42M → 3.89M) next quarter;
avg daily activity 16 h 42 m — sustained institutional adoption.
Beat prob 20 %, miss prob 8 %. Strongest fundamental signal.
→ LONG EV ≈ +45 % Model: trend GBM + analyst beat/miss tails.



Signal: SHORT ▼ [9/10]
EV (long): -42.0% EV (optimal dir): +42.0%
Std: 18.9% Model: crash
Standalone v* = 21% Max net EV = +44.1k

P1: -72.4%
P5: -64.1%
P10: -59.7%
P25: -52.1%
P50: -43.3%
P75: -33.3%
P90: -6.8%
P95: +5.1%
P99: +11.5%

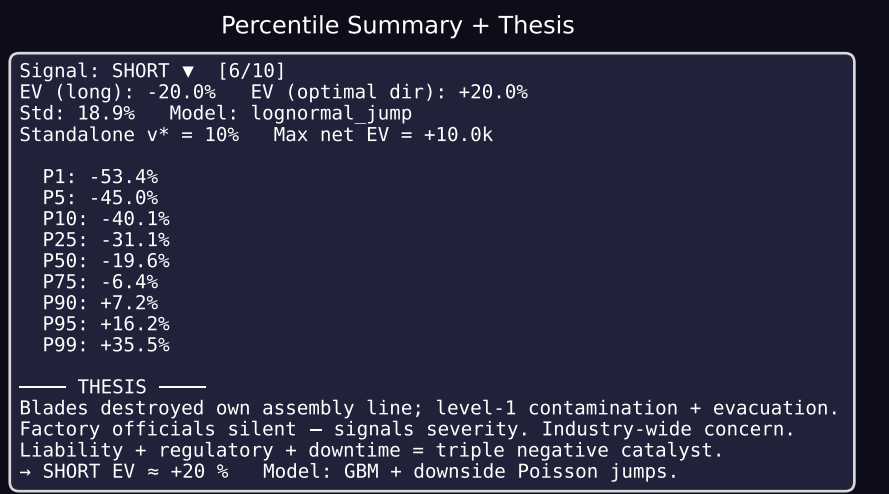
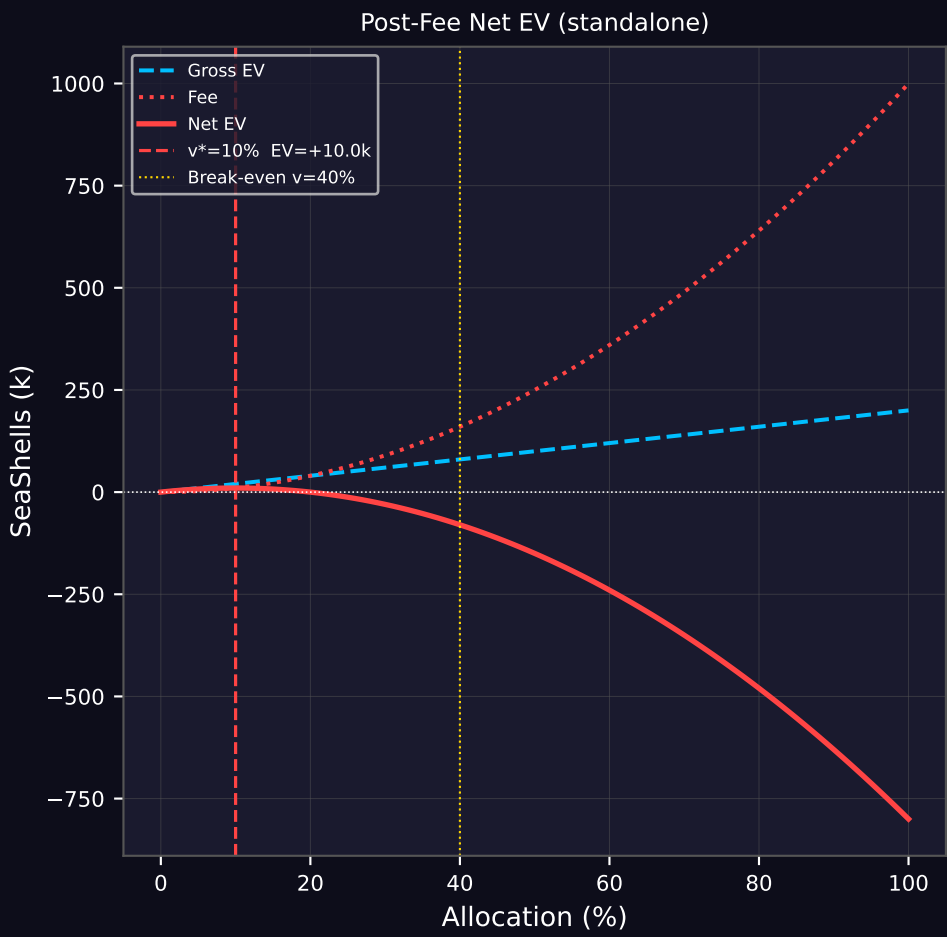
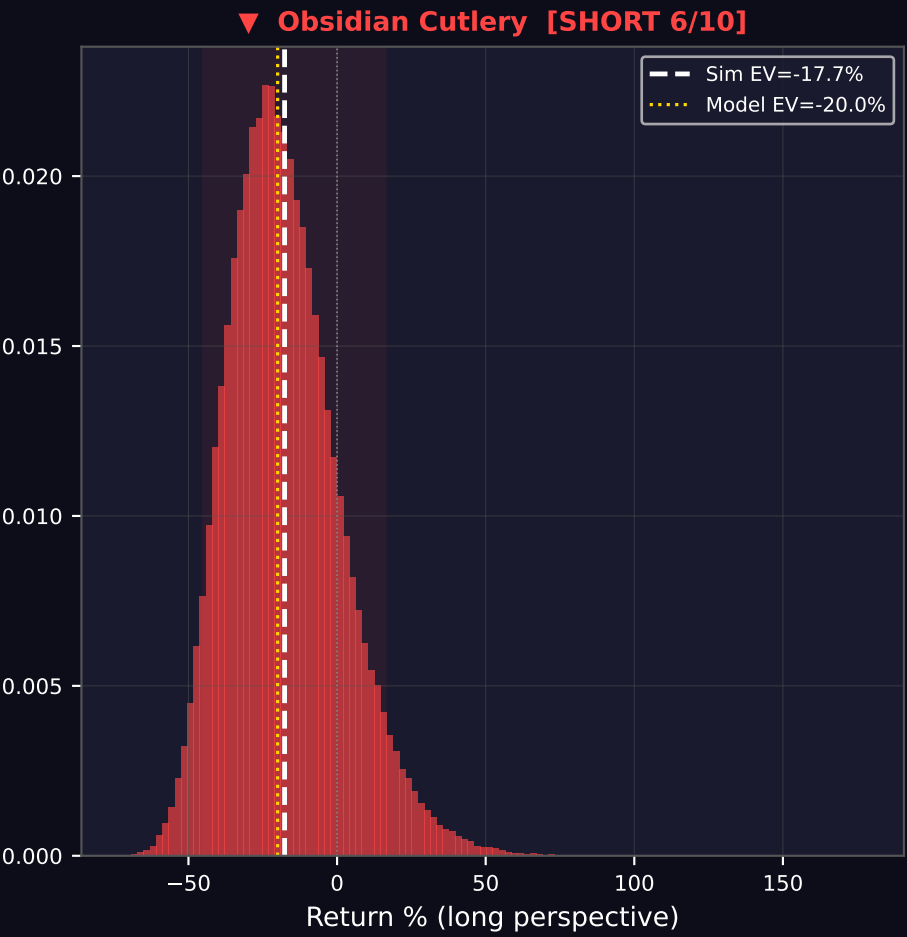
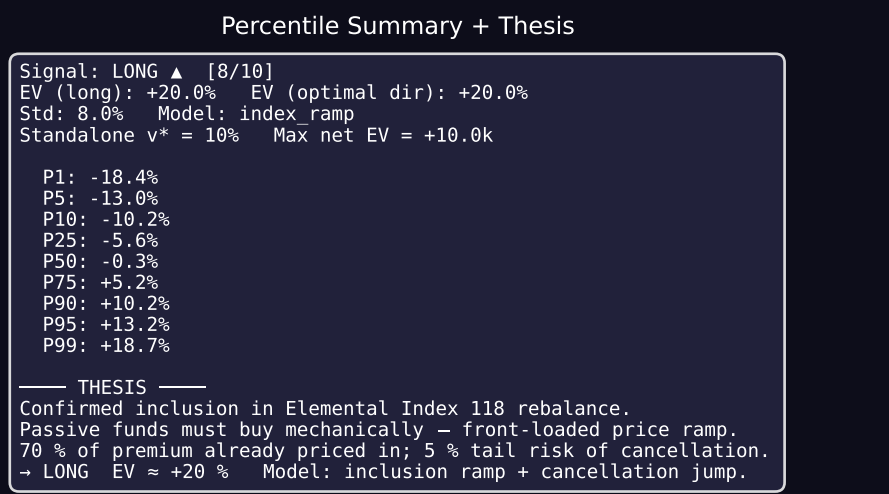
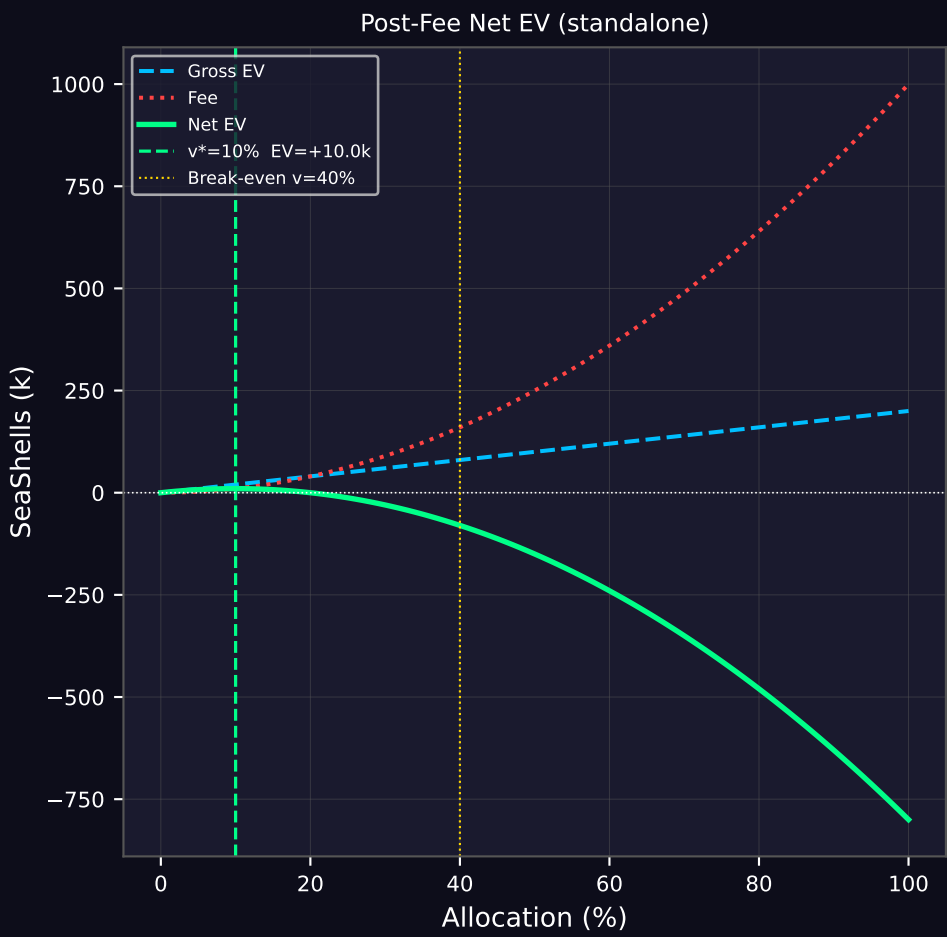
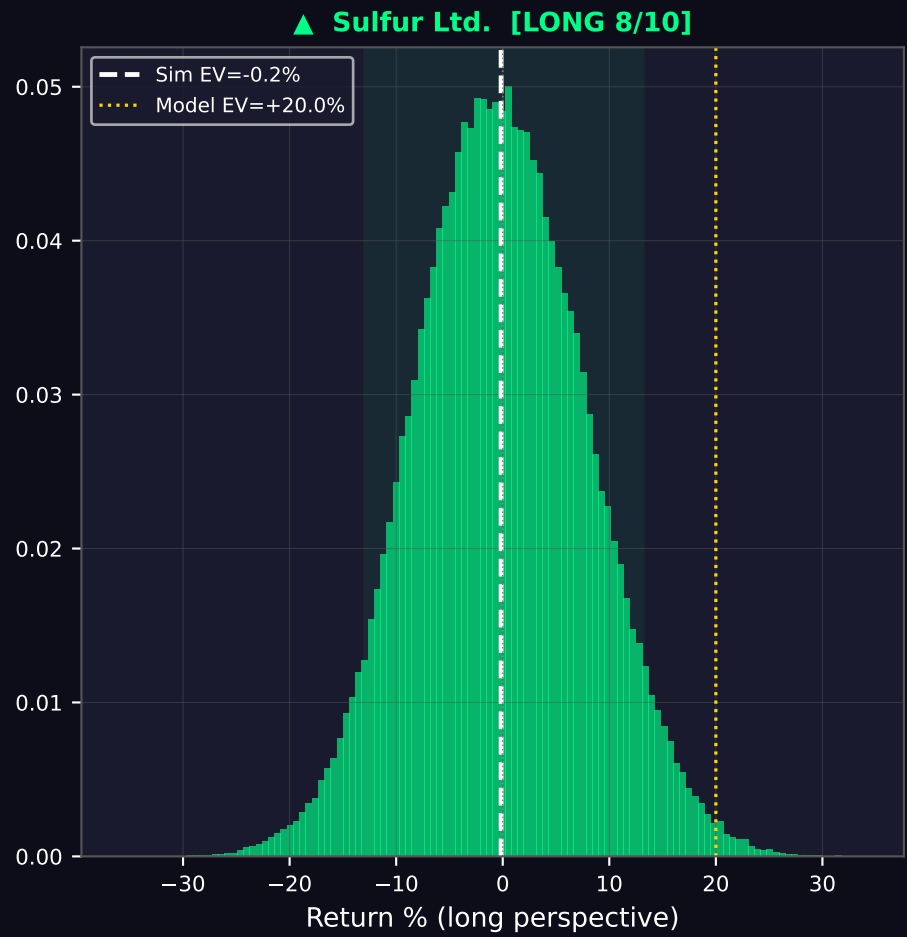
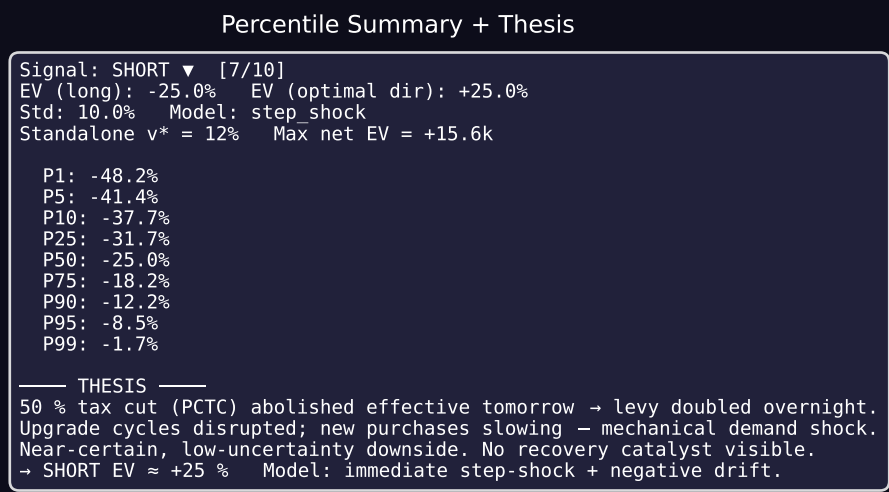
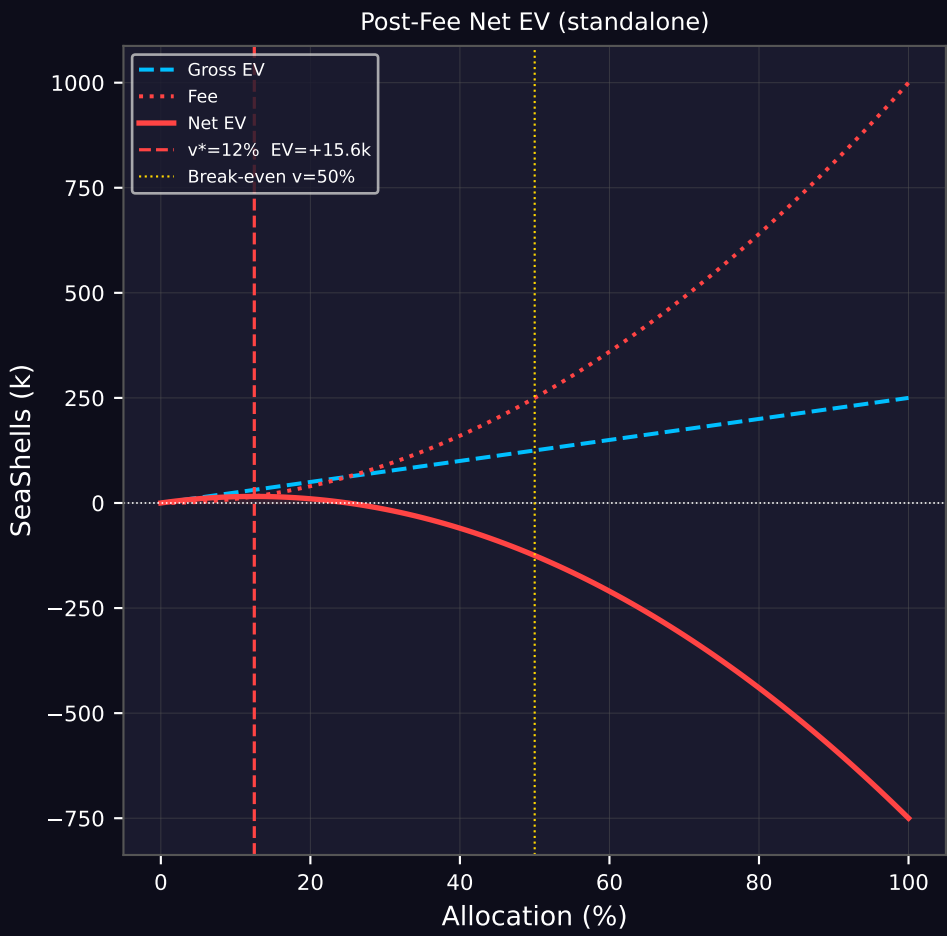
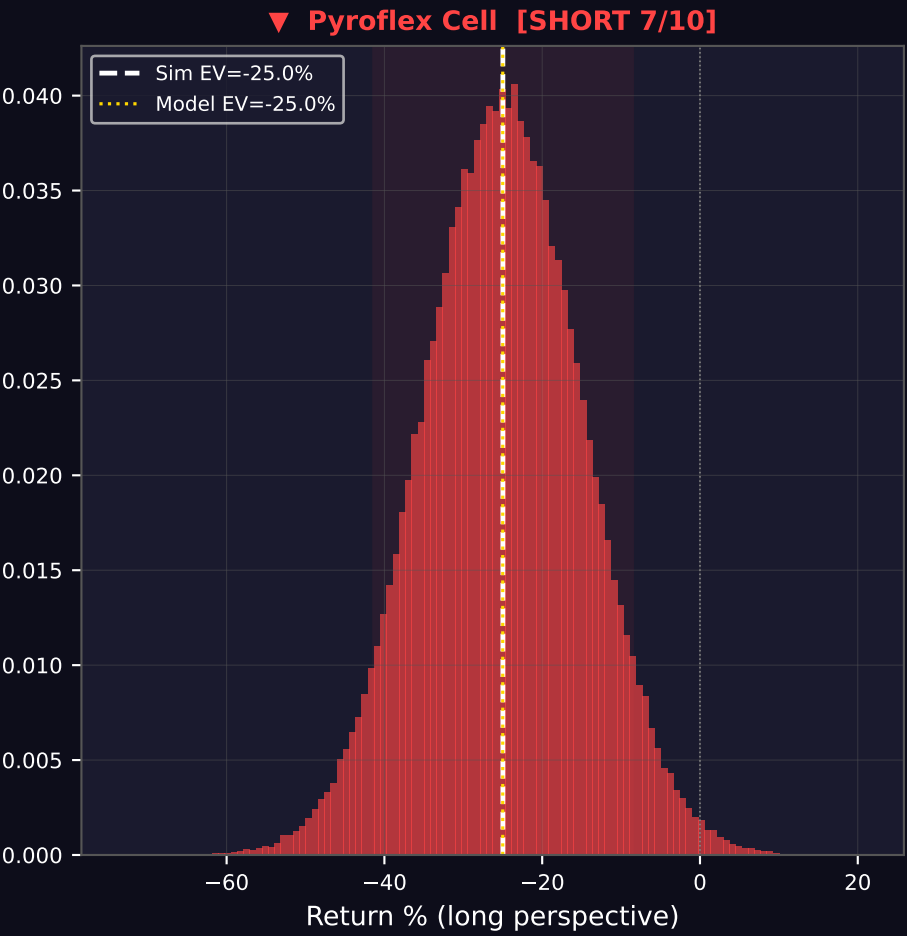
— THESIS —
Actual lava confirmed → health halt, civil lawsuits, vendor returns.
90 % crash probability. Triple hit: regulatory + legal + reputational.
Recovery scenario only 10 % likely (regulators dismiss quickly).
→ SHORT EV ≈ +42 % Model: crash distribution.



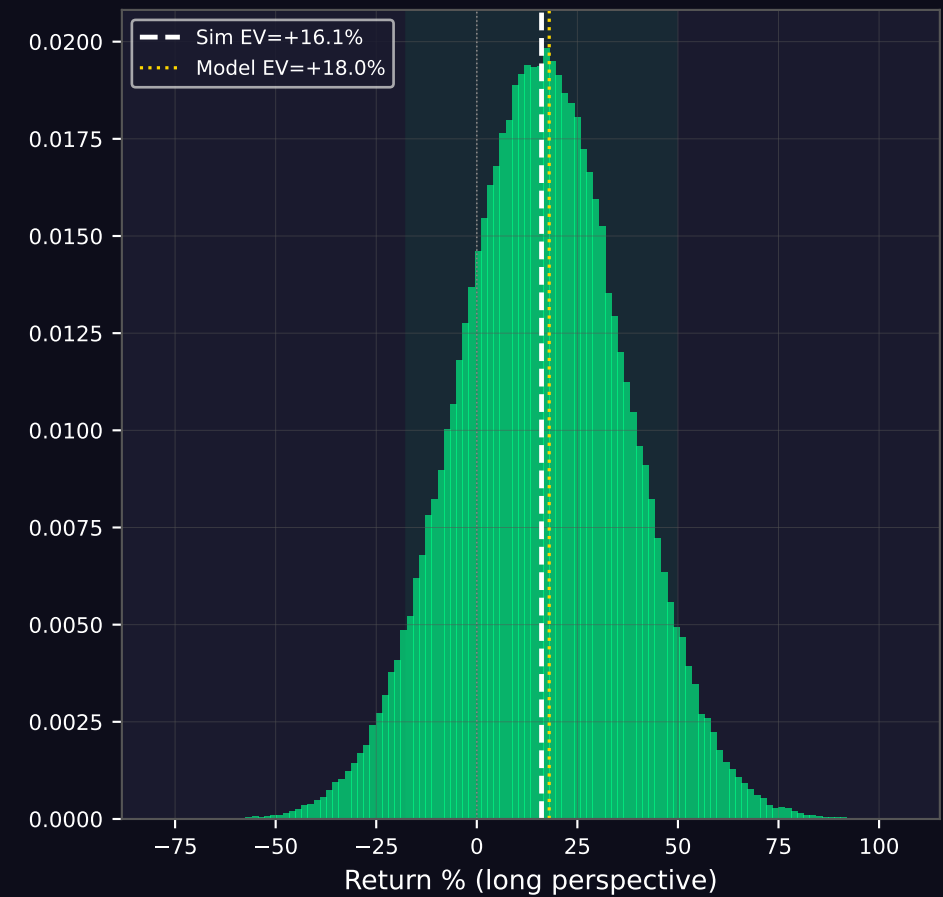
Signal: LONG ▲ [8/10]
EV (long): +30.0% EV (optimal dir): +30.0%
Std: 44.4% Model: lognormal_jump
Standalone v* = 15% Max net EV = +22.5k

P1: -27.5%
P5: -12.6%
P10: -3.1%
P25: +15.4%
P50: +40.5%
P75: +71.1%
P90: +105.1%
P95: +128.8%
P99: +182.1%

— THESIS —
Assembly line destroyed by obsidian blades → manufacturing halt.
Demand spike: Lava Fountain Pen limited-edition, 6-hour queues.
Supply-demand pincer → strong right tail, positive jumps.
→ LONG EV ≈ +30 % Model: GBM + compound Poisson upside jumps.

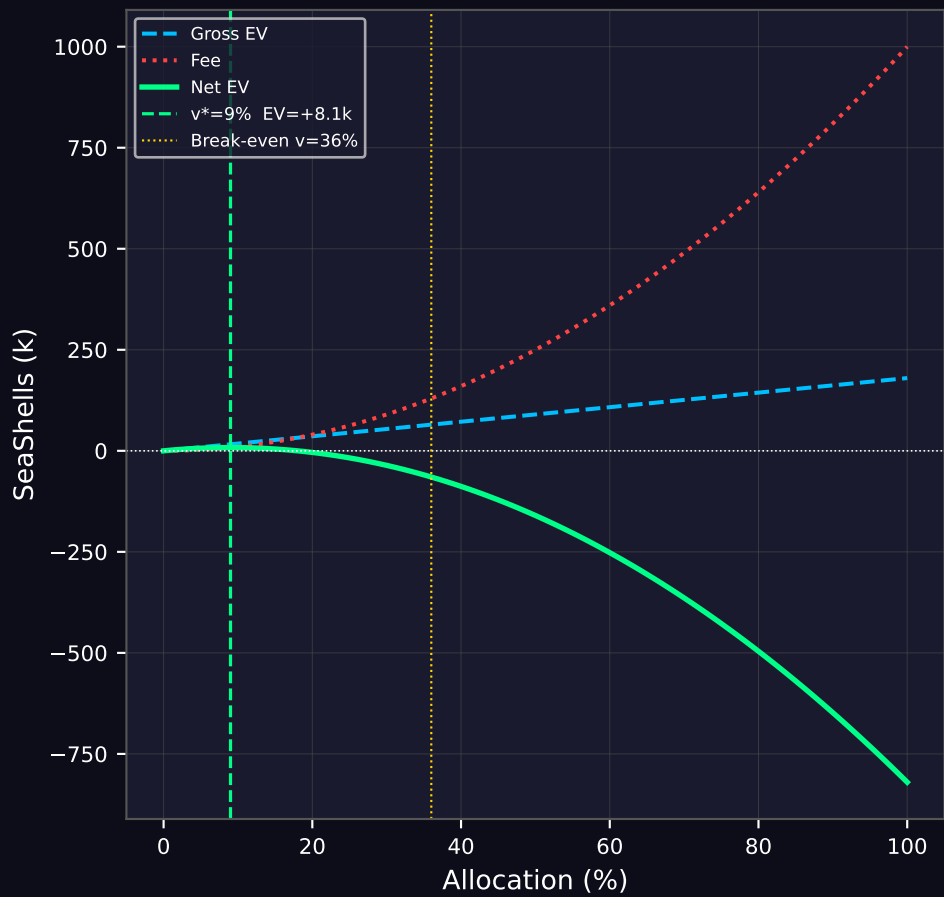


▲ Scoria Paste [LONG 5/10]



Return % (long perspective)

Post-Fee Net EV (standalone)



Allocation (%)

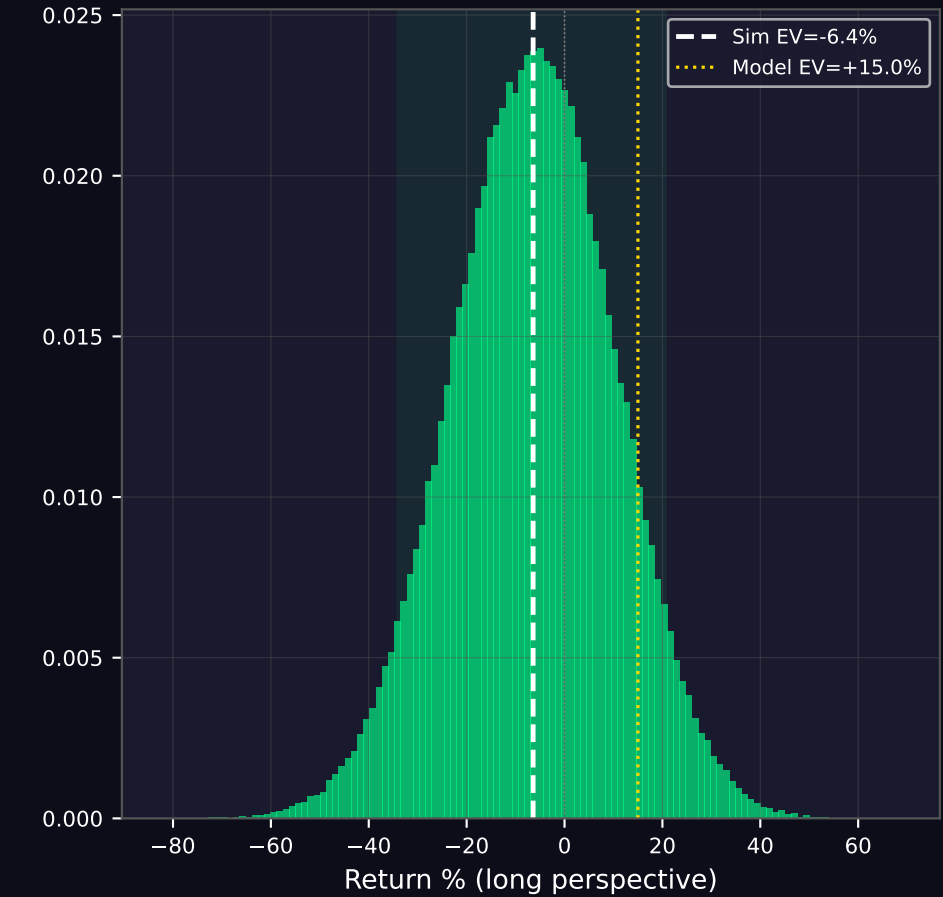
Percentile Summary + Thesis

Signal: LONG ▲ [5/10]
EV (long): +18.0% EV (optimal dir): +18.0%
Std: 20.4% Model: pump decay
Standalone v* = 9% Max net EV = +8.1k

P1: -31.5%
P5: -17.5%
P10: -10.0%
P25: +2.4%
P50: +16.1%
P75: +29.8%
P90: +42.2%
P95: +49.6%
P99: +63.5%

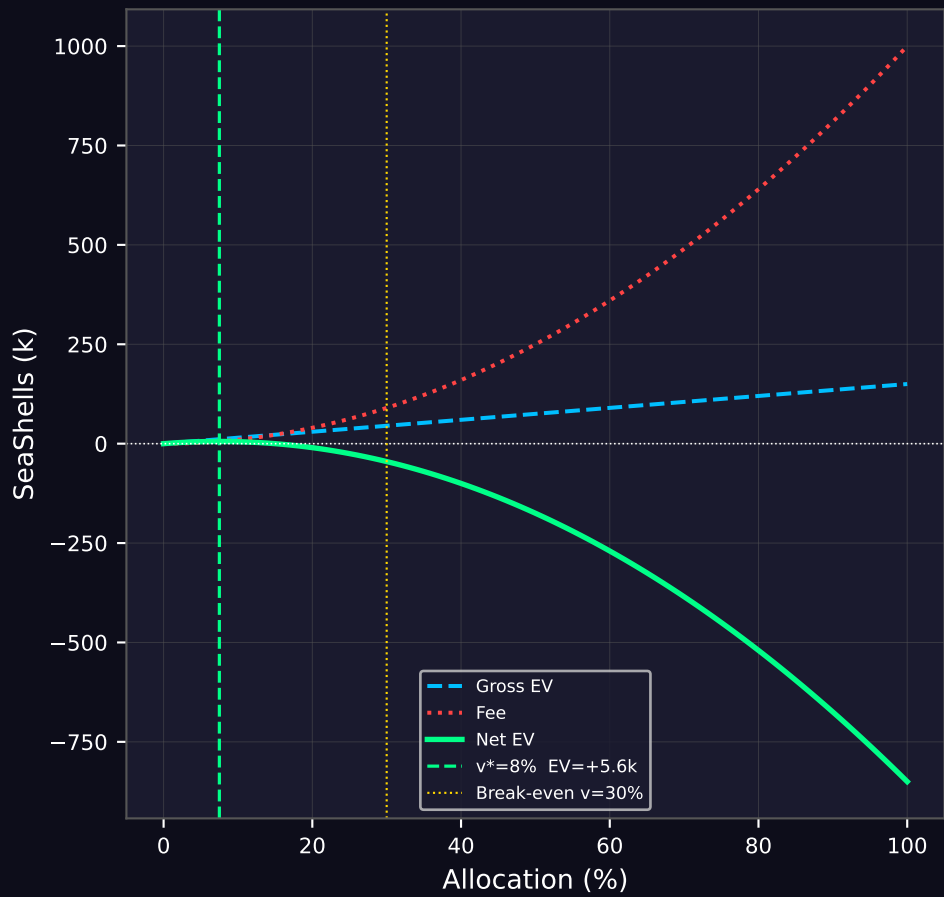
— THESIS —
Lava D. Ray (BrewTube) urges stockpiling – essential infrastructure.
Product called 'the paste that keeps Ignith together'; deep household use.
Pump-decay: initial momentum fades exponentially to fundamental.
→ LONG EV ≈ +18 % Model: pump-decay GBM. Risk: influencer reversal.

▲ Volcanic Incense [LONG 3/10]



Return % (long perspective)

Post-Fee Net EV (standalone)



Allocation (%)

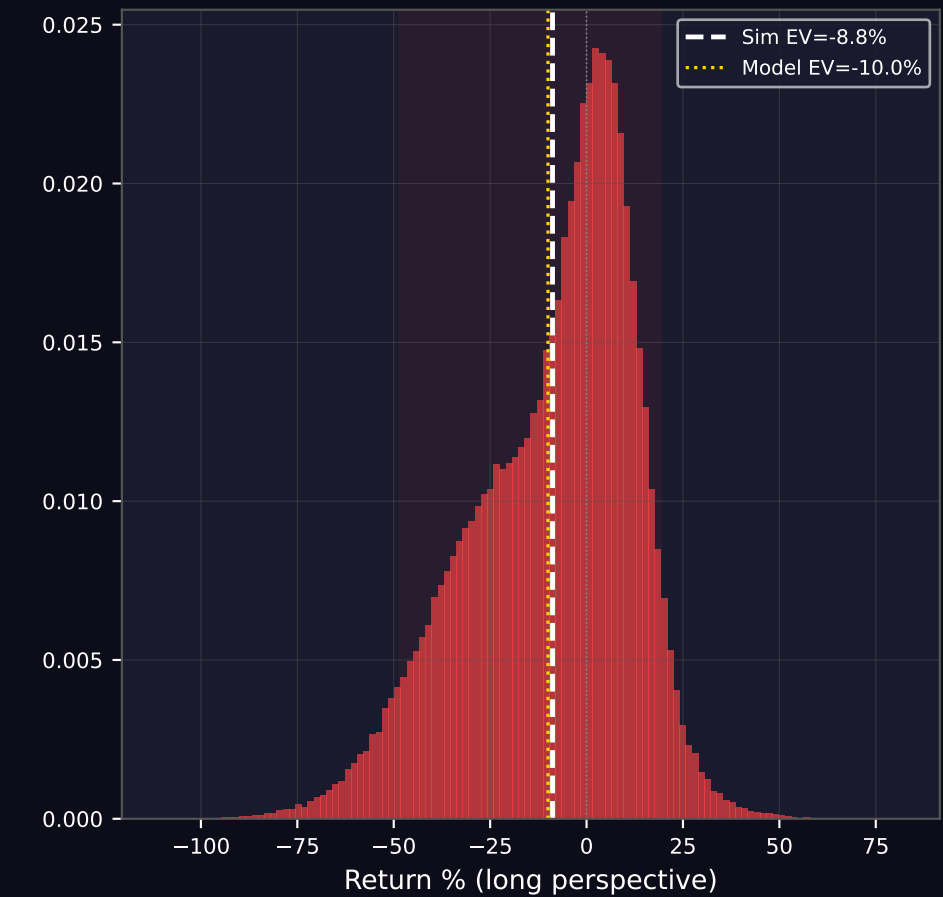
Percentile Summary + Thesis

Signal: LONG ▲ [3/10]
EV (long): +15.0% EV (optimal dir): +15.0%
Std: 16.7% Model: markov
Standalone v* = 8% Max net EV = +5.6k

P1: -45.8%
P5: -34.0%
P10: -27.8%
P25: -17.6%
P50: -6.2%
P75: +4.9%
P90: +14.8%
P95: +20.7%
P99: +31.9%

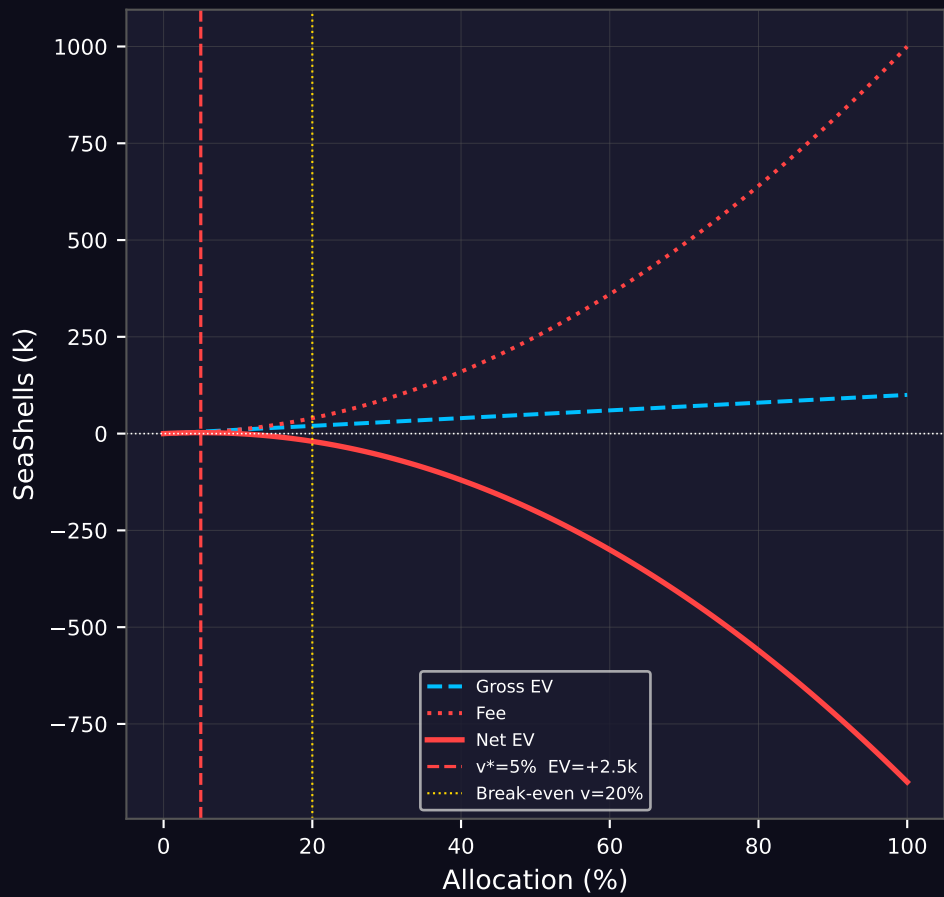
— THESIS —
Nostralico pump: concentrated buying in windows around appearances.
He openly calls for followers to buy – self-reinforcing momentum.
Markov regime-switching: momentum → normal → reversal.
→ LONG EV ≈ +15 % High vol. Risk: 35 % hard reversal. Small alloc.

▼ Ashes of Phoenix [SHORT 5/10]



Return % (long perspective)

Post-Fee Net EV (standalone)



Allocation (%)

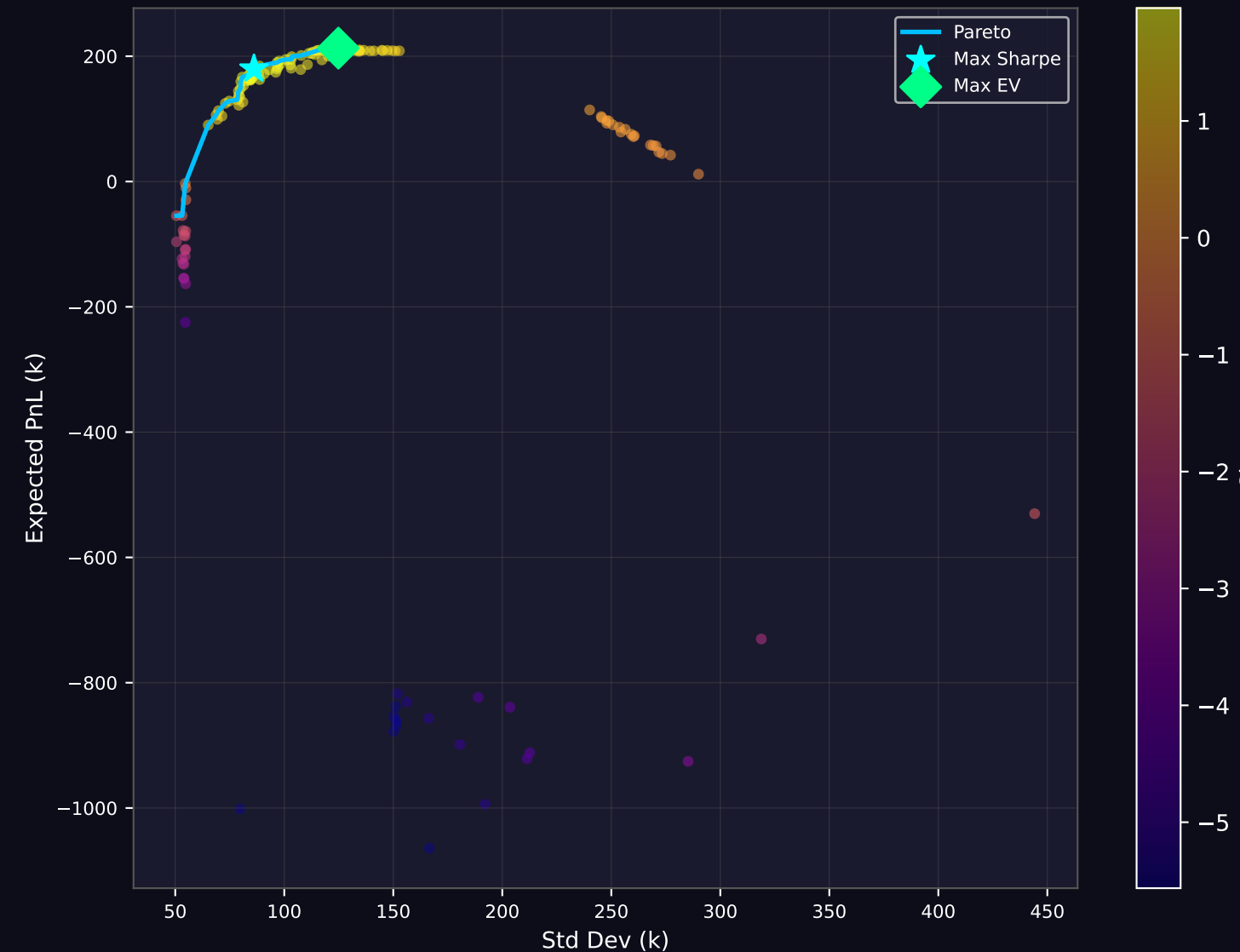
Percentile Summary + Thesis

Signal: SHORT ▼ [5/10]
EV (long): -10.0% EV (optimal dir): +10.0%
Std: 21.3% Model: bimodal
Standalone v* = 5% Max net EV = +2.5k

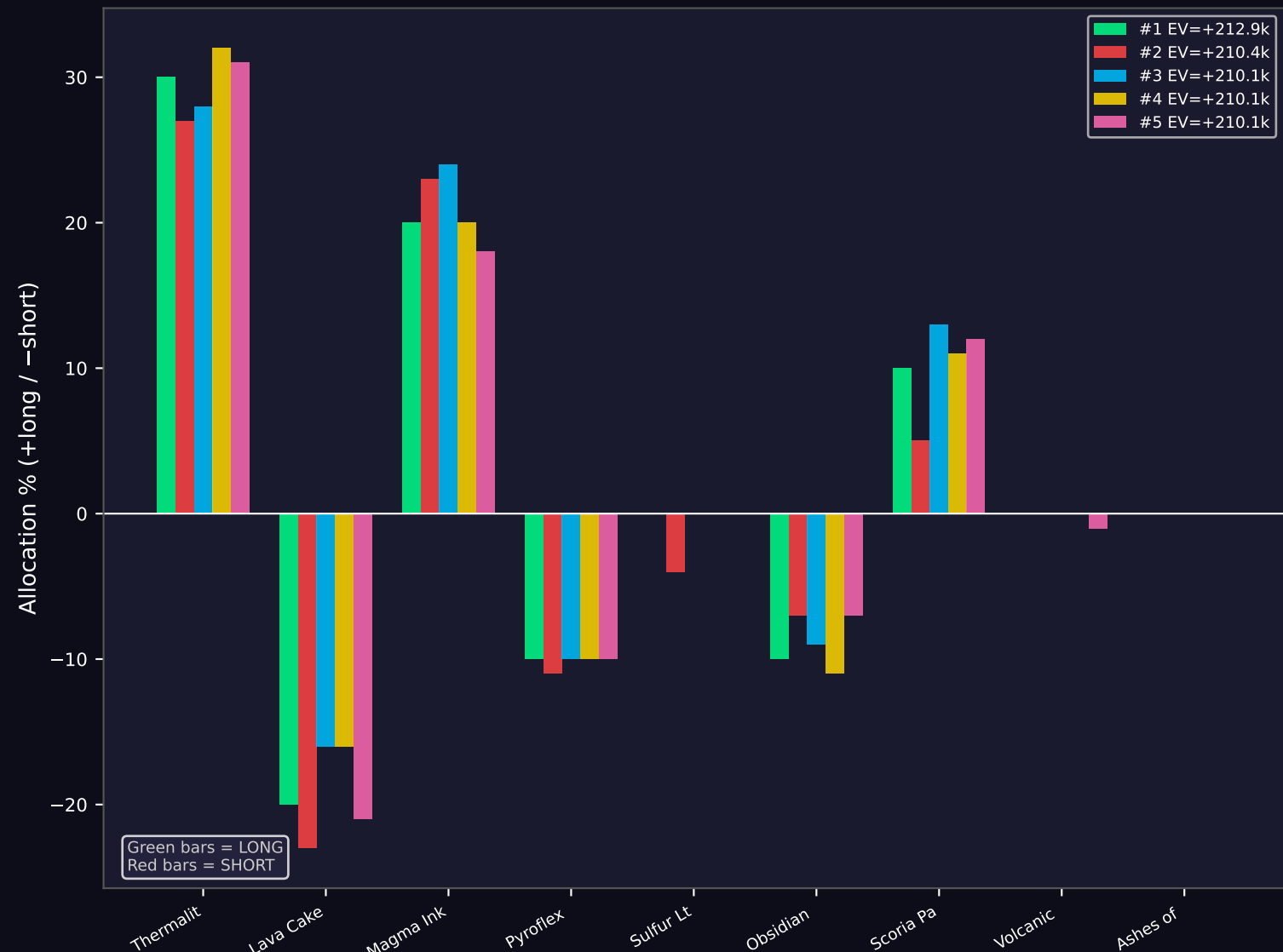
P1: -64.8%
P5: -48.5%
P10: -39.3%
P25: -22.8%
P50: -4.3%
P75: +6.7%
P90: +14.5%
P95: +19.3%
P99: +30.2%

— THESIS —
Viral video: phoenix burned alive for cosmetics ashes – public outrage.
Company counter: 'birds are immortal'. PR survival uncertain.
Bimodal: 40 % PR survives (+5 %), 60 % persistent damage (-18 %).
→ SHORT EV ≈ +10 % High uncertainty. Model: bimodal mixture.

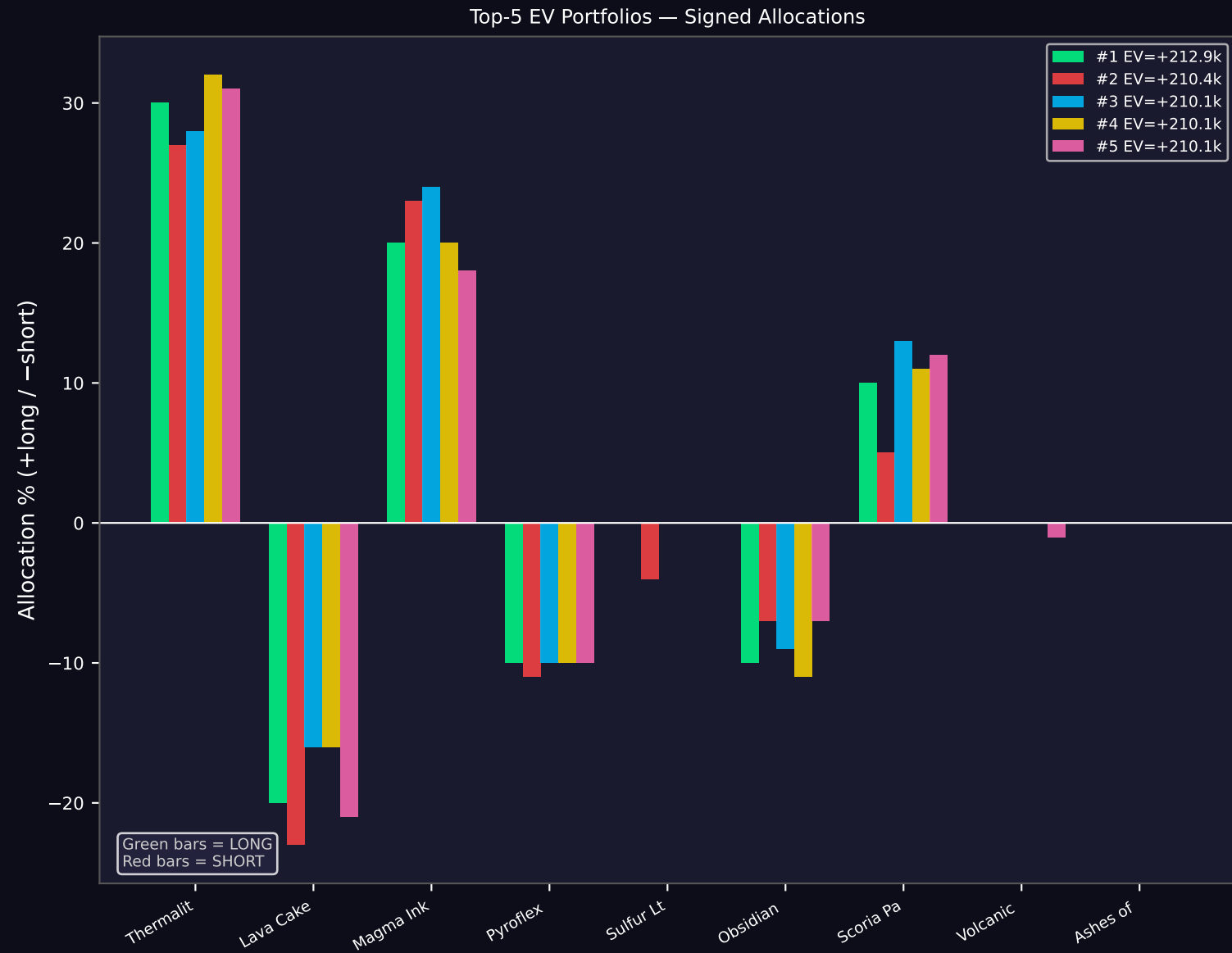
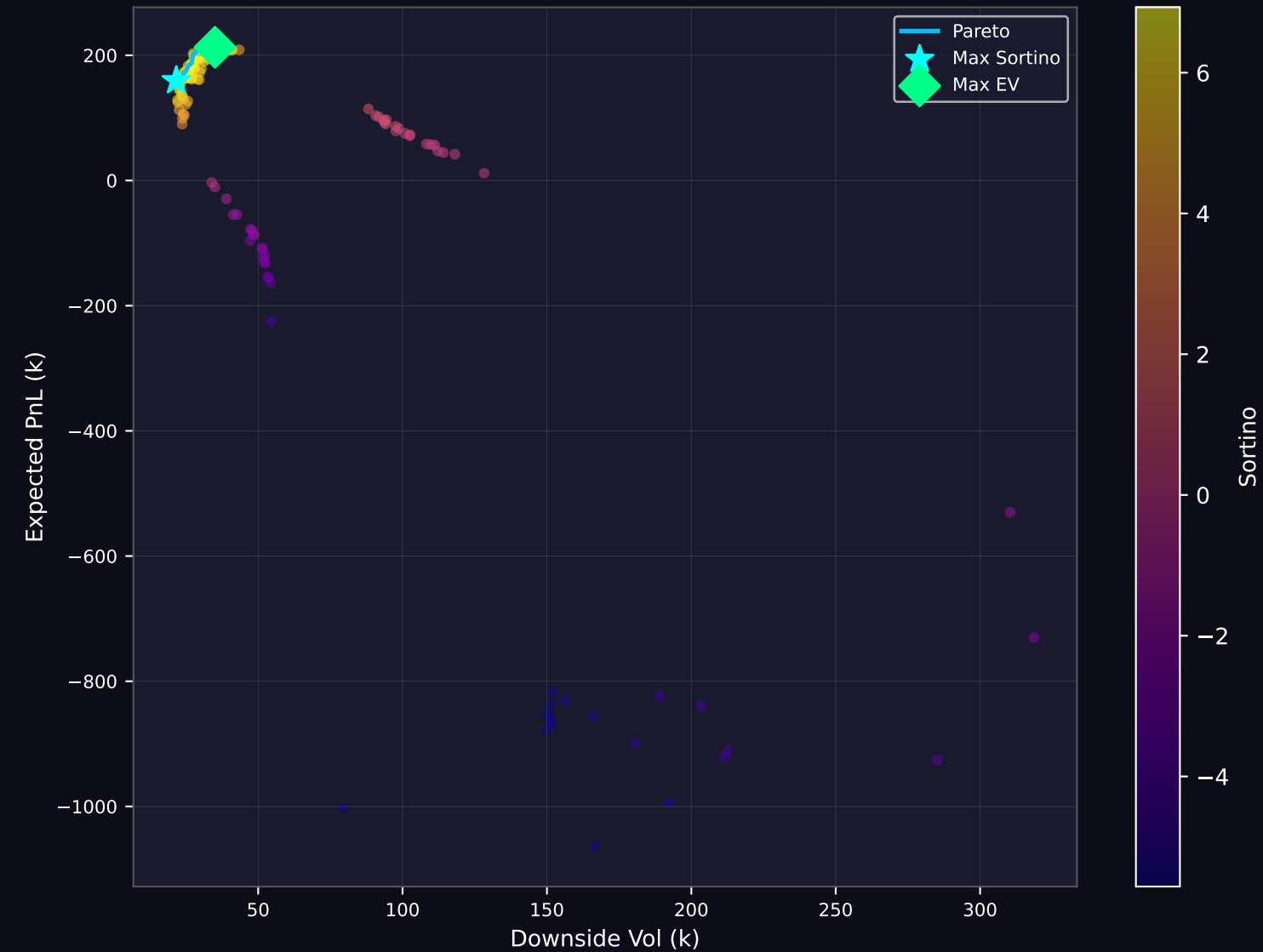
Efficient Frontier — EV vs Std Dev ($\Sigma|\text{alloc}|=100\%$, L+S)



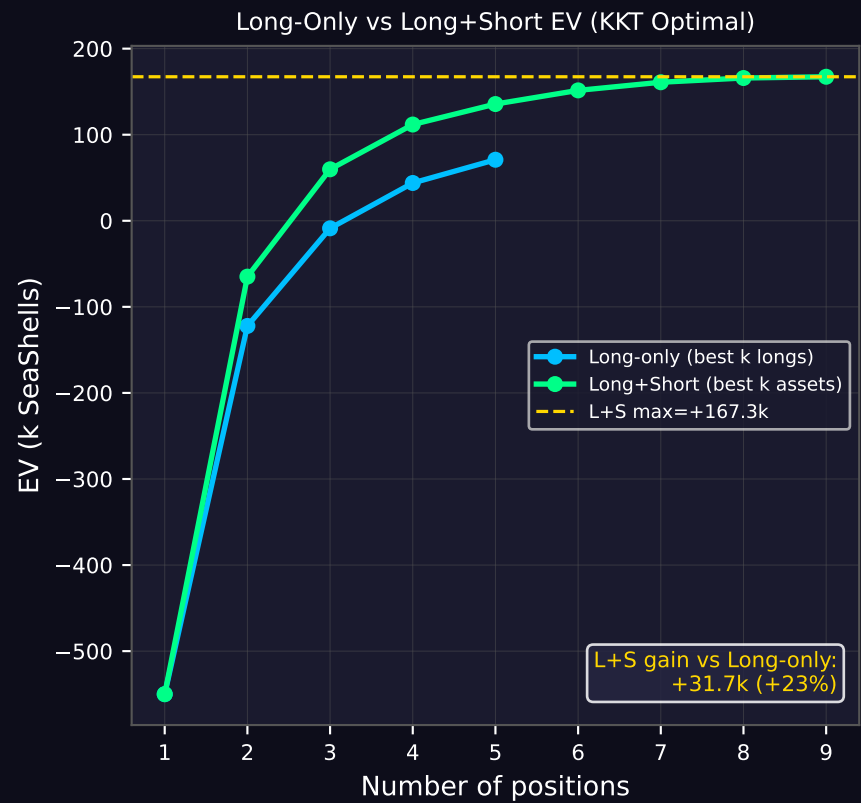
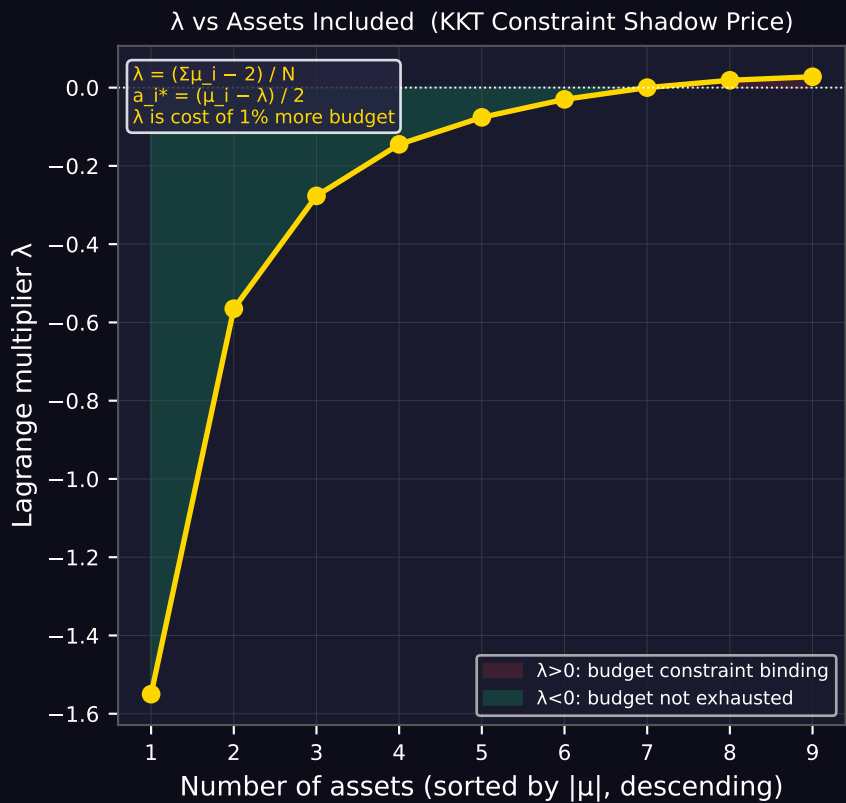
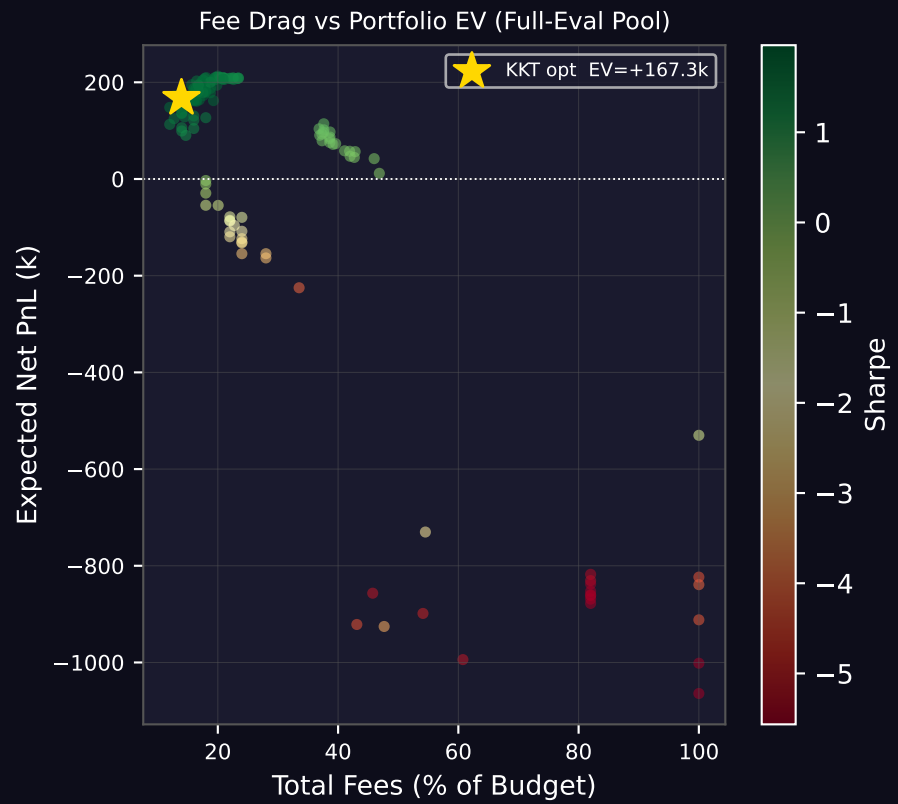
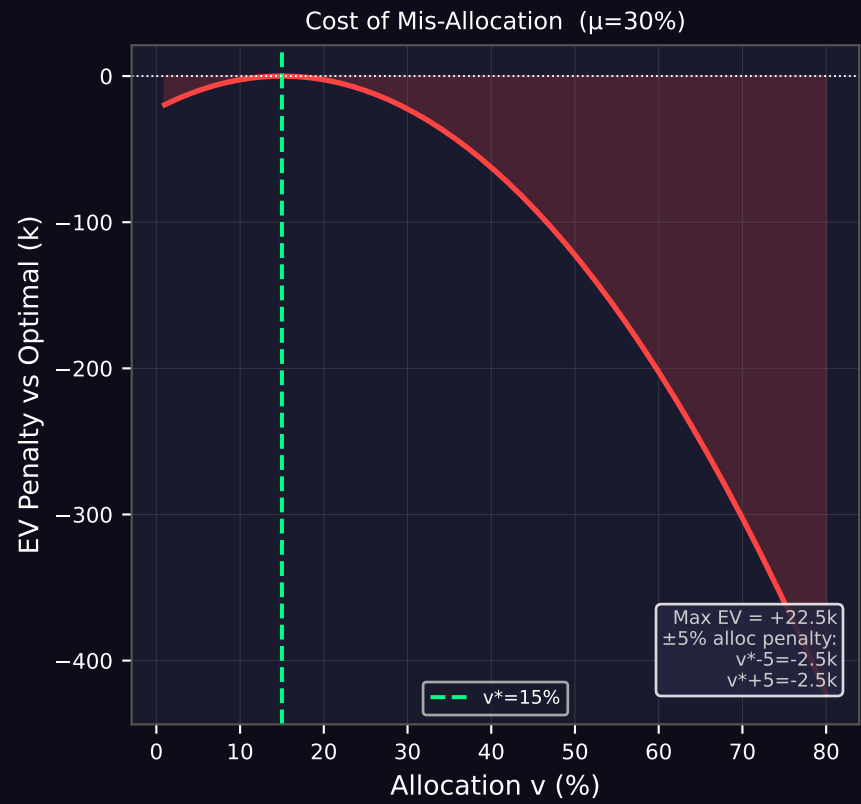
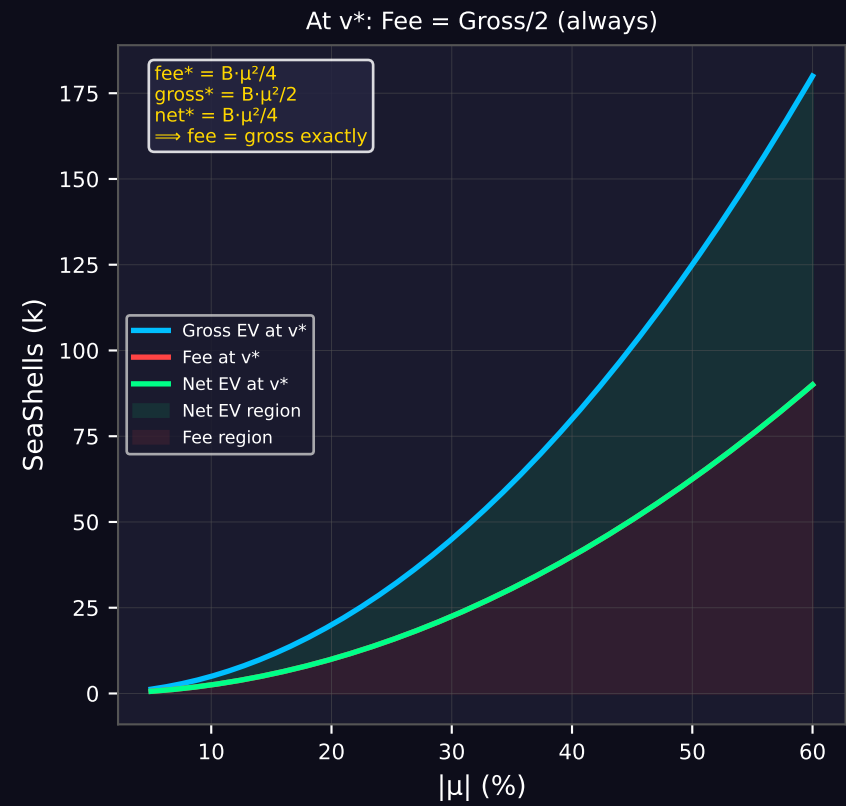
Top-5 EV Portfolios — Signed Allocations



Sortino Frontier — EV vs Downside Vol ($\Sigma|\text{alloc}|=100\%$, L+S)



Fee Optimization Study — Impact of Quadratic Cost on Portfolio Construction



KKT Analytical Allocation Summary

Product	Dir	$ \mu $	v^*_{SA}	v^*_{KKT}	Gross	Fee	Net
Thermalite C	L	45%	22%	21.1%	+95.0k	+44.6k	+50.4k
Lava Cake	S	42%	21%	19.6%	+82.4k	+38.5k	+43.9k
Magma Ink	L	30%	15%	13.6%	+40.8k	+18.5k	+22.3k
Pyroflex Cel	S	25%	12%	11.1%	+27.8k	+12.3k	+15.4k
Sulfur Ltd.	L	20%	10%	8.6%	+17.2k	+7.4k	+9.8k
Obsidian Cu	S	20%	10%	8.6%	+17.2k	+7.4k	+9.8k
Scoria Paste	L	18%	9%	7.6%	+13.7k	+5.8k	+7.9k
Volcanic Inc	L	15%	8%	6.1%	+9.2k	+3.7k	+5.4k
Ashes of Pho	S	10%	5%	3.6%	+3.6k	+1.3k	+2.3k
— TOTAL —					+306.9k	+139.6k	+167.3k

Top 25 Portfolios by EV — Full-Eval (signed: +long / –short)

EV	Median	VaR_95	CVaR_95	Sharpe	Sortino	Win_Rate	Thermal	Lava Ca	Magma I	Pyrofle	Sulfur	Obsidia	Scoria	Volcani	Ashes o
+212.9k	+205.4k	+22.1k	-20.2k	1.71	6.07	96.9%	+30	-20	+20	-10	+0	-10	+10	+0	+0
+210.4k	+201.1k	+13.9k	-30.2k	1.61	5.71	96.2%	+27	-23	+23	-11	-4	-7	+5	+0	+0
+210.1k	+199.7k	+10.0k	-32.1k	1.56	5.86	95.9%	+28	-16	+24	-10	+0	-9	+13	+0	+0
+210.1k	+202.1k	+17.8k	-23.9k	1.66	6.09	96.6%	+32	-16	+20	-10	+0	-11	+11	+0	+0
+210.1k	+203.6k	+23.2k	-19.0k	1.74	5.98	97.0%	+31	-21	+18	-10	+0	-7	+12	-1	+0
+209.7k	+202.5k	+32.6k	-6.1k	1.81	6.86	97.7%	+29	-16	+18	-13	+0	-6	+13	-3	-2
+209.6k	+198.9k	+6.5k	-38.3k	1.54	5.49	95.6%	+26	-23	+25	-11	-3	-8	+3	+0	-1
+209.5k	+197.5k	-4.1k	-49.8k	1.45	5.17	94.6%	+27	-22	+27	-6	+1	-8	+7	-1	-1
+209.3k	+199.1k	+13.0k	-29.3k	1.59	5.92	96.2%	+26	-20	+24	-18	+0	-6	+6	+0	+0
+209.2k	+196.9k	-8.2k	-52.8k	1.42	5.27	94.2%	+31	-17	+27	-13	+0	-10	+2	+0	+0
+208.9k	+203.8k	+29.2k	-13.4k	1.82	6.05	97.3%	+28	-24	+17	-11	-1	-11	+7	+0	-1
+208.8k	+197.1k	-13.8k	-61.4k	1.40	4.81	93.7%	+31	-23	+27	-7	+0	-6	+6	+0	+0
+208.7k	+196.3k	-5.1k	-50.0k	1.44	5.28	94.5%	+29	-20	+27	-8	-3	-5	+2	-2	-4
+208.7k	+201.5k	+21.4k	-20.7k	1.71	6.04	96.8%	+31	-21	+19	-15	+0	-2	+4	-2	-6
+208.6k	+193.6k	-12.4k	-56.4k	1.36	5.25	93.7%	+26	-17	+30	-12	+0	-7	+7	-1	+0
+208.5k	+198.9k	-4.0k	-50.3k	1.48	5.10	94.6%	+33	-22	+24	-12	-1	-7	+1	+0	+0
+208.3k	+197.3k	+1.4k	-42.5k	1.50	5.51	95.1%	+30	-19	+25	-17	+0	-2	+4	+0	-3
+208.3k	+200.5k	+29.2k	-10.4k	1.76	6.55	97.5%	+25	-18	+20	-14	+0	-6	+14	+0	-3
+208.2k	+195.0k	-13.5k	-58.8k	1.38	5.08	93.6%	+31	-17	+28	-12	+0	-2	+9	+0	-1
+208.2k	+198.3k	+6.2k	-36.2k	1.55	5.69	95.6%	+31	-15	+23	-10	+0	-7	+14	+0	+0
+208.0k	+199.4k	+5.6k	-41.3k	1.56	5.21	95.5%	+28	-26	+23	-13	+0	-5	+3	-2	+0
+208.0k	+199.2k	+3.8k	-40.1k	1.55	5.48	95.4%	+36	-18	+21	-14	+0	-4	+7	+0	+0
+206.6k	+201.8k	+29.7k	-12.6k	1.83	6.08	97.4%	+29	-23	+16	-12	+0	-13	+3	-1	-3
+206.6k	+201.3k	+25.9k	-15.7k	1.79	6.06	97.2%	+33	-21	+15	-13	+0	-8	+10	+0	+0
+205.7k	+199.7k	+19.6k	-23.6k	1.71	5.71	96.7%	+30	-23	+18	-15	+0	-2	+11	-1	+0

Best Portfolio Deep-Dive (Full-Eval, signed L/S allocations)

